

## Fushuai Jiang

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| CONTACT INFORMATION  | Department of Mathematics<br>City University of Hong Kong<br>83 Tat Chee Ave, Kowloon Tong<br>Kowloon, Hong Kong                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <a href="mailto:fusjiang@cityu.edu.hk">fusjiang@cityu.edu.hk</a><br><a href="https://jfsmath.github.io/">https://jfsmath.github.io/</a>            |
| RESEARCH             | Real Analysis, Functional Analysis, Optimal Transport, Machine Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                    |
| ACADEMIC EMPLOYMENT  | 2026-Current                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>City University of Hong Kong, Assistant Professor</b>                                                                                           |
|                      | 2023-2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>University of Maryland, Novikov Postdoctoral Fellow</b>                                                                                         |
|                      | 2022 Fall                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>ICERM, Postdoctoral Fellow</b>                                                                                                                  |
| EDUCATION            | <b>University of California, Davis</b><br>Ph.D. in Mathematics, June 2022<br>Dissertation Topic: Nonnegative Smooth Interpolation<br>Advisor: Kevin Luli<br>M.S. in Mathematics, June 2018                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                    |
|                      | <b>University of California, Los Angeles</b><br>B.S. in Mathematics, June 2017<br>Departmental Highest Honors, Cum Laude Latin Honors                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                    |
| HONORS AND AWARDS    | 2021-2022                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <a href="#">Yueh-Jing Lin Fund.</a><br>Awarded to high-achieving mathematics students.                                                             |
|                      | 2021-2022                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <a href="#">Graduate Research Award.</a><br>Annual internal fellowship awarded to promising graduate researchers.                                  |
|                      | 2020                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <a href="#">Alice Leung Scholarship.</a><br>Awarded to graduate student with exceptional promise in mathematical research.                         |
|                      | 2020-2021                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <a href="#">Summer Graduate Student Researcher (GSR) Award.</a><br>In recognition of exceptional achievements in our graduate education community. |
| JOURNAL PUBLICATIONS | <ol style="list-style-type: none"><li>15. F. Jiang<br/><a href="#">Roots, trace, and extendability of flat nonnegative functions</a><br/><b>International Mathematics Research Notices</b>, online, 2024</li><li>14. F. Jiang, C. Liang, Y. Liang, and G.K. Luli<br/><a href="#">Univariate Range-Restricted <math>C^2</math> Interpolation Algorithms</a>,<br/><b>Journal of Computational and Applied Mathematics</b>, 425:115040, 2023.</li><li>13. C. Fefferman, F. Jiang, G.K. Luli<br/><a href="#"><math>C^2</math> interpolation with range restriction</a>,<br/><b>Revista Matemática Iberoamericana</b>, 39(2):649–710, 2023</li><li>12. F. Jiang, G.K. Luli, K. O'Neill<br/><a href="#">Smooth selection for infinite sets</a>,<br/><b>Advances in Mathematics</b>, 407:108566, 2022.</li></ol> |                                                                                                                                                    |

11. F. Jiang, G.K. Luli, K. O'Neill,  
[On the shape fields finiteness principle](#)  
**International Mathematics Research Notices**, 23:18895–18918, 2022.
10. F. Jiang, G.K. Luli,  
[Algorithms for nonnegative  \$C^2\(\mathbb{R}^2\)\$  interpolation.](#)  
**Advances in Mathematics**, 385:107756, 2021.
9. F. Jiang, G.K. Luli,  
 [\$C^2\(\mathbb{R}^2\)\$  nonnegative extension by bounded-depth operators.](#)  
**Advances in Mathematics**, 375:107391, 2020.
8. F. Jiang, G.K. Luli,  
[Nonnegative  \$C^2\(\mathbb{R}^2\)\$  interpolation.](#)  
**Advances in Mathematics**, 375:107364, 2020.
7. K. Xu, F. Jiang, W. Zhang, and Y. Hao  
[Micromachined integrated self-adaptive nonlinear stops for mechanical shock protection of MEMS.](#)  
**Journal of Micromechanics and Microengineering**, 28:064006, 2018.
6. F. Jiang and G.K. Luli  
[Optimal  \$C^{1,1}\$  and Quasi-Optimal  \$C^2\$  Monotone Interpolation with Curvature Control](#)  
[arXiv:2605.14302](#)
5. R. Balan, F. Jiang  
[Factorization of positive-semidefinite operators with absolutely summable entries](#)  
[arXiv:2409.20372](#).
4. J. Chhoa, M. Ivanitskiy, F. Jiang, S. Li, D. McBride, T. Needham, and K. O'Hare  
[Metric properties of partial and robust Gromov-Wasserstein distances](#)  
[arXiv:2411.02198](#).
3. F. Jiang  
[Nonnegative Whitney extension problem for  \$C^1\(\mathbb{R}^n\)\$ .](#)  
[arXiv:1912.06327](#).
2. K. Xu, N. Zhu, F. Jiang, W. Zhang, and Y. Hao  
[A Transfer Function Approach to Shock Duration Compensation for Laboratory Evaluation of Ultra-High-G Vacuum-Packaged MEMS Accelerometers.](#)  
**IEEE 32nd International Conference on Micro Electro Mechanical Systems (MEMS)**, 676–679, 2019.
1. K. Xu, N. Zhu, F. Jiang, W. Zhang, and Y. Hao  
[Micromachined integrated shock protection via a self-adaptive nonlinear system.](#)  
**19th International Conference on Solid-State Sensors, Actuators and Microsystems (TRANSDUCERS)**, 524–527, 2017.

## TALKS

|            |                                                                                                                                                                 |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01/09/2025 | <i>Efficient algorithm for non-negative smooth interpolation</i><br><a href="#">AMS Special Session on Adversarial, Interpretable, and Explainable AI</a>       |
| 11/21/2024 | <i>Finding a smooth solution to an underdetermined system</i><br><a href="#">One World Mathematics of Information, Data, and Signals (1W-MINDS) Seminar</a>     |
| 10/19/2024 | <i><math>\ell_1</math>-squared summable series and a problem by Feichtinger</i><br><a href="#">Frame Theory Days 2024</a>                                       |
| 10/11/2024 | <i>How to find a smooth solution to an underdetermined system?</i><br>Johns Hopkins postdoc seminar                                                             |
| 07/06/2024 | <i>Smooth selection of convex sets</i><br><a href="#">Loo-Keng Hua Lecture</a> , Chinese Academy of Sciences                                                    |
| 06/24/2024 | <i>Partial optimal transport</i><br><a href="#">MRC 2024: Mathematics of Adversarial, Interpretable, and Explainable AI</a>                                     |
| 03/24/2024 | <i>Primal and dual optimization problems related to matrix factorizations</i><br><a href="#">AMS Special Session on Bases and Frames in Hilbert Spaces, III</a> |
| 08/14/2023 | <i>Positivity and the Whitney Extension Problem</i><br><a href="#">The 15th Whitney Problems Workshop</a>                                                       |
| 06/18/2023 | <i>Recent development in Whitney extension problems with constraint</i><br>Shenzhen University                                                                  |
| 03/10/2023 | <i>Quasi-optimal <math>C^2</math> interpolation with range restriction</i><br>CUNY GC Harmonic Analysis and PDE <a href="#">Seminar</a>                         |
| 10/14/2022 | <i>Smooth Selection of Convex Sets</i><br><a href="#">Harmonic Analysis and Convexity Program</a> , ICERM                                                       |
| 10/06/2022 | <i>Quasi-optimal <math>C^2</math> interpolation with range restriction</i><br>Fall Fourier Talks, University of Maryland at College Park                        |
| 07/08/2021 | <i>Recent Results on Whitney extensions with constraints</i><br><a href="#">The 14th Whitney Problems Workshop</a>                                              |
| 05/15/2020 | <i>Efficient algorithm for nonnegative <math>C^2</math> interpolation</i><br>UC Davis PDE and Applied Math <a href="#">Seminar</a>                              |
| 02/23/2020 | <i>A gentle introduction to pseudodifferential calculus</i><br>UC Davis Student-run Analysis & PDE <a href="#">Seminar</a>                                      |
| 10/15/2019 | <i>Fitting smooth functions to data - Whitney's problems and beyond</i><br>UC Davis Student-run Research <a href="#">Seminar</a>                                |
| 05/11/2019 | <i>Whitney extension problem and interpolation of data</i><br><a href="#">Bay Graduate Math Conference</a>                                                      |
| 02/08/2019 | <i>Interpolation of data by nonnegative <math>C^2</math> functions</i><br>UC Davis PDE and Applied Math <a href="#">Seminar</a>                                 |

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|--------------------|-----------|-----------|--------------------------------------------------------------------|
| TEACHING           | Fall      | 2025      | Data Visualization and Presentation (R)                            |
|                    | Spring    | 2025      | Applied Probability and Statistics (senior level, R)               |
|                    | Fall      | 2024      | Data Visualization and Presentation (R)                            |
|                    | Spring    | 2024      | Data Science and Machine Learning (Python)                         |
|                    | Spring    | 2024      | Applied Probability and Statistics (senior-level, R)               |
|                    | Fall      | 2023      | Calculus III                                                       |
|                    | Spring    | 2023      | Linear Algebra and Applications (MATLAB)                           |
|                    | Summer    | 2022      | Real Analysis                                                      |
|                    | Fall      | 2020      | Teaching Assistant, Real Analysis                                  |
|                    | Spring    | 2020      | Teaching Assistant, Graduate Analysis                              |
|                    | Summer    | 2019      | Instructor, Calculus II                                            |
|                    | Spring    | 2019      | Teaching Assistant, Graduate Analysis                              |
|                    | Various   | 2017-2021 | Teaching Assistant, Calculus I-III                                 |
| SERVICE            | 2024      |           | <i>Assistant organizer</i> , Mathematics Research Communities, AMS |
|                    | 2021-now  |           | <i>Mentor</i> , Undergraduate research                             |
|                    | 2023      |           | <i>Organizer</i> , The 15th Whitney Problems workshop              |
|                    | 2021      |           | <i>Organizer</i> , The 14th Whitney Problems workshop              |
|                    | 2021      |           | <i>Organizer</i> , 2021 Davis Math Conference                      |
|                    | 2021      |           | <i>Organizer</i> , Student Run Analysis&PDE Seminar                |
|                    | 2021-2022 |           | <i>Mentor</i> , Direct Reading Project                             |
|                    | 2018-2020 |           | <i>Tutor</i> , STEM Café at Women's Resources and Research Center  |
|                    | 2019-2020 |           | <i>Vice President</i> , Galois group (graduate student group)      |
| PROGRAMMING SKILLS | 2018-2020 |           | <i>Representative</i> , Graduate Student Association               |
|                    | R         |           | (dplyr, ggplot, Shiny)                                             |
|                    | Python    |           | (NumPy, SciPy, scikit-learn, pandas)                               |
|                    | MATLAB    |           |                                                                    |